

Abstract Algebra I

Exercise Sheet 4: Cosets

Equivalence Relations

Question 1. For this question $\mathbb{N}$ denotes the natural numbers including zero. Decide whether the following are equivalence relations:

- for $a, b \in \mathbb{Z}$ let $a \sim b$ if $a = b + 3k$ for some $k \in \mathbb{Z}$;
- for $a, b \in \mathbb{Z}$ let $a \sim b$ if $a = b + 1 + 3k$ for some $k \in \mathbb{Z}$;
- for $a, b \in \mathbb{Z}$ let $a \sim b$ if $a = b + 2 + 3k$ for some $k \in \mathbb{Z}$;
- $\leq$ on $\mathbb{Z}$;
- for $a, b \in \mathbb{N}$ let $a \sim b$ if $\gcd(a, b) = 1$;
- for $a, b \in \mathbb{Z}$ let $a \sim b$ if $|a| = |b|$;
- for $a, b \in \mathbb{R}$, $a, b > 0$, let $a^2 \sim b^2$ if $a - b \in \mathbb{Z}$;

Cosets

Question 2. For the following groups and subgroups compute all the left and right cosets. Determine whether the subgroup is normal. Decide whether the groups and subgroups are simple (whether it has non trivial normal subgroups).

- $G = \mathbb{Z}_6$ and $H = \mathbb{Z}_3$.
- $G = \mathbb{Z}$ and $H = 2\mathbb{Z}$.
- $G = \mathbb{Z}_9$ and $H = \langle 3 \rangle$.
- $G = \mathbb{Z}_2 \times \mathbb{Z}_3$ and $H = \langle (0, 1) \rangle$.
- $G = \mathbb{Z} \times \mathbb{Z}$ and $H = \langle (1, 0), (0, 2) \rangle$.
- Let G be the group of symmetric transformations of a regular pentagon. Let H be the subgroup of rotations of the pentagon.
- Let G be the group of symmetric transformations of a regular pentagon. Let H be any subgroup generated by a single flip of the pentagon.

Normal subgroups

Question 3. Decide whether the following subgroups are normal:

- $A_3 < S_3$;
- $A_4 < S_4$;
- $A_5 < S_5$;
- $\langle \rho \rangle < S_5$ for any $\rho \in S_5$ (check for each type of permutation);

Question 4. Find the normal subgroups of the group of symmetric transformations of a square and a cube.